Factor-base designs: m = 4 longer-budget cells L1 and L2 (run-20261019T1633Z)

Kind: solver-axis stage diagnostic (AGENTS.md §2). Observations only. Nothing here prices a pipeline, reports S or shows a speedup. The protocol is `PREREGISTRATION.md` Amendment 5, registered before any L1 or L2 row (commit `1551b4dc0`). The analysis is `runs/analysis-amendment5.txt` (native `analyze --amendment5`). Every number below is copied from it or from the row files.

Run

- **Commands** (`B=target/release/examples/fb_designs_slope`, `R=research/factor_base_designs_20261015/runs`, one system at a time, L1 before L2):
  ```
  $B run --arms A0 --m 4 --t-min 4 --t-max 4 --budget 14400 --grace 600 --vmem-kb 14500000 --out-dir $R --file 'L1-{arm}-m4-t4'
  $B run --arms A11 --m 4 --t-min 5 --t-max 5 --budget 14400 --grace 600 --vmem-kb 14500000 --out-dir $R --file 'L2-{arm}-m4-t5'
  $B analyze $R --amendment5 > $R/analysis-amendment5.txt
  ```
- **Binaries:** L1 ran on `81577f519`. That is the Amendment 5 harness plus one recorded field, `peak_rss_kb`: the child is reaped with `wait4`, and its `ru_maxrss` is recorded as descriptive only. L2 ran on `4aec76a5a`. Between the two commits, only `group_of` and the new `analyze --amendment5` changed, not the run path. Both harness versions reproduce the committed `analysis-amendment2.txt` and `analysis-amendment4.txt` byte for byte. A smoke run at A0 m = 2, t = 2..3 reproduced the committed `field_ops` (2131, 40982).
- **Host:** "vm", Intel Xeon @ 2.80 GHz, 4 cores, a Claude Code cloud container. `MemTotal` is 16 479 424 kB, which is at least 15 GB, so `--vmem-kb` stayed at 14 500 000 and no Amendment 6 was needed. The host was otherwise idle while each system ran. Note that C2 ran on a 2.10 GHz host, which affects wall time only.
- **Window (UTC, 2026-10-19):** L1 16:33:42–16:51:18. L2 16:52:33–16:55:58.
- **Limits as registered:** budget 14 400 s, grace 600 s (hard kill at 15 000 s), `ulimit -v 14500000`.
- **Records:** `runs/L1-A0-m4-t4.{jsonl,log}`, `runs/L2-A11-m4-t5.{jsonl,log}`, `runs/L2-A11-m4-t5.dmesg.txt` (kernel log lines copied verbatim), `runs/analysis-amendment5.txt`.

L1: A0, m = 4, t = 4 (b = 4, |V| = 16, 18 variables, 19 equations)

system	status	time (s)	peak RSS (kB)	D	field_ops	limit
target 0	aborted	516.7	12 835 532	—	—	malloc failed under `ulimit -v 14500000` (SIGABRT)
target 1	aborted	539.7	12 891 360	—	—	malloc failed under `ulimit -v 14500000` (SIGABRT)

- **Identity with C2:** both L1 systems match C2′s A0 systems on cell seed (18358879315), target seeds (18358879571, 18358879827), `n_vars` = 18, `n_eqs` = 19, planted x, x_R, curve and set (2 of 2).
- C2 ran the same systems under `ulimit -v 13000000`, and they aborted at 536.9 s and 512.8 s. With 1.5 GB more address space, L1 aborted at about the same wall time (516.7 s, 539.7 s). No partial F4 values exist for a SIGABRT. The only partial value recorded for this cell is F4′s earlier 300 s timeout: degree 7 reached, `max_cols` 102 516 (open marker in the figure).

L2: A11, m = 4, t = 5 (b = 5, |V| = 32, 42 variables, 43 equations)

system	status	time (s)	peak RSS (kB)	D	field_ops	limit
target 0	aborted	97.3	13 974 324	—	—	SIGKILL: kernel memory-cgroup OOM
target 1	aborted	108.0	13 974 336	—	—	SIGKILL: kernel memory-cgroup OOM

- The harness recorded `child exited: signal: 9 (SIGKILL)`. The harness sends SIGKILL only at its 15 000 s hard limit, so it did not send this one. The kernel log (`runs/L2-A11-m4-t5.dmesg.txt`) shows the memory-cgroup OOM killer on the container's cgroup, at `memory: usage 14007508kB, limit 14007508kB`. Kill times (boot + 2639.98 s and 2748.17 s, i.e. 16:54:10Z and 16:55:57Z) and pids (10420, 10700) match the two systems.
- **Deviation:** the binding limit for L2 was this cgroup limit, 14 007 508 kB of charged memory. That is below the registered `ulimit -v 14500000`. The registered `MemTotal` check (16 479 424 kB) did not reveal it. L1 is not affected: its systems hit `ulimit -v` with peak RSS 12.8 GB, below the cgroup limit.

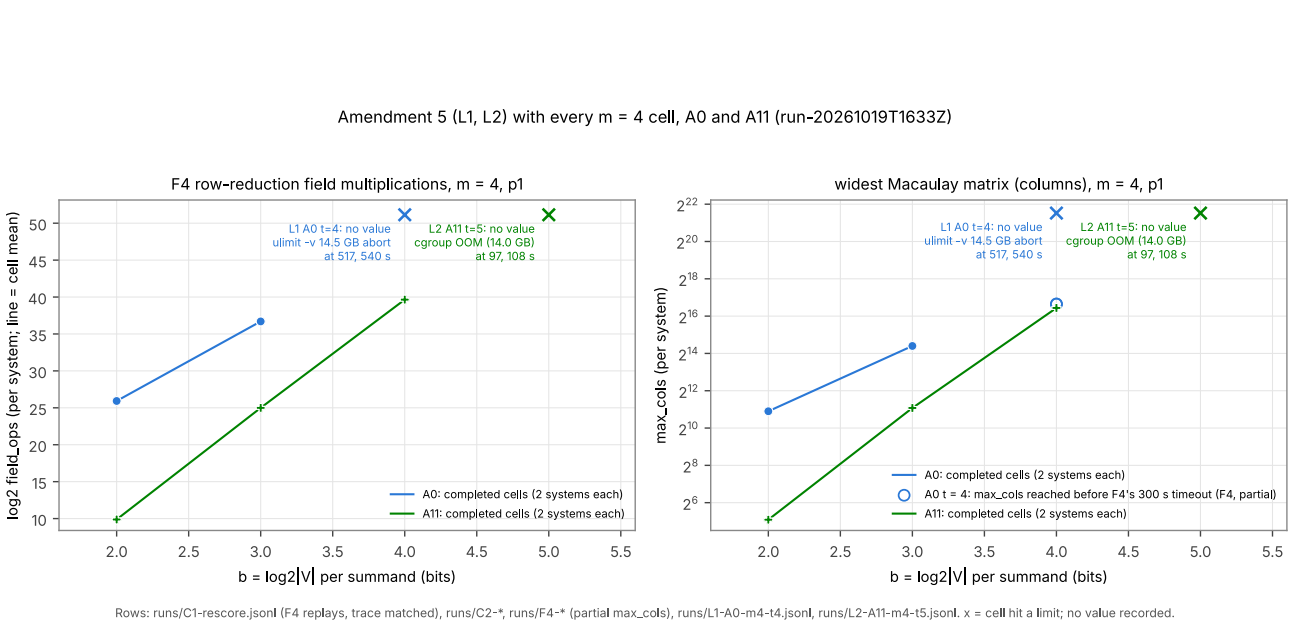


Figure 1: $\log_2$ `field_ops` (left) and `max_cols` (right, $\log_2$ scale) against b at m = 4, p_1, for A0 and A11. Points are recorded systems (2 per cell), and lines join the cell means of completed cells. × marks a cell that hit a limit, with no value recorded (L1, L2). The open circle is the `max_cols` reached by A0 at t = 4 before F4′s 300 s timeout (`runs/F4-A0-m4.jsonl`), a partial value, not the cell's maximum. Rows: `runs/C1-rescore.jsonl` (F4 replays, trace matched), `runs/C2-*`, `runs/F4-*`, `runs/L1-*`, `runs/L2-*`. Run run-20261019T1633Z, `plot.py`.

σ and κ at m = 4 (completed cells only)

series	completed cells	κ	m − 1	σ	verdict
A0 m=4 p_1 (F4 via C1; L1 t = 4 not completed)	2 (t = 2, 3)	10.773	3	1.0000	inconclusive (fewer than 3 cells)
A11 m=4 p_1 (F4 via C1, C2 t = 4; L2 t = 5 not completed)	3 (t = 2, 3, 4)	14.886	3	1.5000	κ: cost-unfavourable; σ: no middle ground (σ ≥ 0.5)

Per-system values in the completed cells (`field_ops`): A0 t = 2: 63 703 685, 63 704 352. A0 t = 3: 111 465 267 887, 111 465 240 139. A11 t = 2: 941, 941. A11 t = 3: 33 417 634, 33 402 245. A11 t = 4 (C2): 863 032 546 892, 863 026 754 688.

Decision rules (Amendment 5)

- **R1, does A11′s constant factor survive at m = 4, t = 4?** L1 did not complete. Outcome: **A0 not measurable at t = 4 within 14.5 GB and 4 h** (limit: `ulimit -v` at 517 s and 540 s). No cost comparison is made.
- **R2, A11 at m = 4, t = 5.** L2 did not complete (limit: kernel memory-cgroup OOM at 14 007 508 kB, about 100 s per system). A11′s 3-cell verdicts stand: κ = 14.886 (cost-unfavourable, κ ≥ m − 1 = 3), σ = 1.5000.
- A0′s m = 4 series gained no third cell, so its σ and κ verdicts remain undecided.

Predictions scorecard

prediction	observed	outcome
L1: A0 aborts for memory again	both systems aborted for memory (`ulimit -v 14500000`)	held
L2: A11 does not complete within 14.5 GB	did not complete; the kernel memory cgroup killed both systems at 14 007 508 kB, below the registered 14.5 GB	not decided as registered

Limits and what is missing

- No `field_ops`, D or `max_cols` exist for A0 at m = 4, t = 4, or for A11 at m = 4, t = 5. The only partial values are wall time and peak RSS. A memory abort leaves no F4 counters.
- L2 never had the registered 14.5 GB. Whether A11 at t = 5 completes between 14.0 and 14.5 GB, or within 4 h on a larger host, is not measured here. The independent lab runner is still unavailable from this container (Amendment 4, C2).
- Two systems per cell, one host, no isolation receipt. Wall times are descriptive only.